
CS 4080-5080: Reinforcement Learning

Home Work Assignment 2

Out: 03-17-2026, Due 04-09-2026

This assignment requires you to implement a reinforcement learning agent from scratch, i.e., you will not use any reinforcement learning library or framework. You can use any numeric processing library in the language you have chosen. You can also use any library that implements a neural network. Do not use any library that implements Q-learning or deep Q-learning directly. You can program in any language of your choice.

Consider a simplified “parking lot” with 8 parking spots marked P , and one entrance marked E . It is described in terms of a 6×7 grid, with a single, as in Figure 1. There is barrier to movement between the two rows of parking.

A car agent can move only forward, backward, or sideways. For a simple car, we make the assumption that there is no angular steering. A car should not drive over parking spots on its way to a goal parking spot.

The agent needs to learn a policy or a number of policies that will take it from the entrance to any of the parking spots. Assume all grid cells are of equal size and a car fits well inside any of the cells. We assume that the car is a square and the grid cells are squares as well.

1. *Traditional or Tabular Q-learning*: Here, you program a reinforcement learning agent using table(s).
 - (a) The parking lot diagram given in the figure is a starting point. You can make changes if you like to make it better represent an actual parking lot, within the constraints of the homework. What you do in this assignment is left intentionally open.

How would you represent the various components that are required to program the car to learn to park in any of the parking spots? Discuss each component. Explain how you model the barriers, and how you make sure the agent does not fall off the edges.
 - (b) You are also asked to implement Q-learning algorithm and the Double Q-learning variant of it.

Write the algorithms you implement in terms of pseudocode and briefly explain with reference to lines in the pseudocode. Discuss how the base algorithm is improved by the second algorithm.
2. *Deep Q-learning*: You are now going to implement Deep Q-learning for the same problem.
 - (a) What are the inputs and the outputs of the neural network? What is the neural network going to learn?

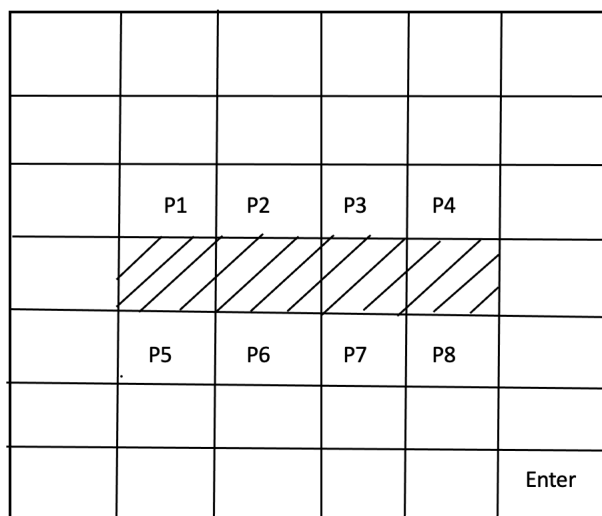


Figure 1: A Simplified Parking Lot

What is the architecture of the neural network you are going to build? Why did you choose this architecture?

Are you going to use a replay memory? What is the structure of the replay memory? Why have you chosen this structure? How big is the replay memory? How are you going to fill the replay memory? You can play with sizes of the replay memory and make observations regarding effects.

- (b) Implement the algorithm using any programming language of choice. Run the algorithm several times. Show one or more of the policies learned. Change any relevant parameters and rerun a few times. Comment on how changes in parameter values change learning. It is a good idea to include graphs or tables for such analysis.
 - (c) Perform one or more enhancements. The problem as given is trivial. You can make it more complex in terms of size of the environment or other details, and adapt your solution. Discuss how you make the problem a little more complex, and what kinds of adaptations you have to perform.
3. Write a short paper giving details of what you have done, the results you have obtained, the problems you have faced, and how you have overcome the problems. Use the format you use for your semester project papers. The maximum number of pages is 4 for content, followed by an extra page for references, if necessary.
 4. Extra credit may be given based on substantial additional work.

You do not need to hand in soft copies of your programs for this assignment. Written answers

in the paper to questions given or issues raised above are sufficient. You will have to demo your program the week it is due or before. Please upload copy of your paper on Canvas on the due date.